

The Prompt Ladder

1. Version 1:
 - a. Weak baseline prompt
 - b. Build the backend for an AI chatbot.
2. Version 2:
 - a. Layer added: Defined audience.
 - b. Build the backend for an AI chatbot for university students (current, alumni, prospective).
 - c. Notes:
 - i. Changes in the prompt: the target audience is now defined.
 - ii. Improvement in the output: the backend's design was now centered around university-related information and questions instead of a generic chatbot.
 - iii. Still failed: the responses are still very generic.
 - iv. Next improvement: Real context.
3. Version 3:
 - a. Layer added: Real context.
 - b. Build the backend for a voice-based AI chatbot for university students (current, alumni, prospective). The chatbot is being developed as an AI-powered university assistant for an internship capstone project. It must support both voice and text conversations, retrieve information from the university website using Retrieval-Augmented Generation (RAG), provide citations with every response, and maintain a low response time.
 - c. Notes:
 - i. Changes in the prompt: the project's functionality is described.
 - ii. Improvement in the output: the output became tailored to the project by incorporating embeddings, vector databases, LiveKit integration, citation handling, and latency optimisation.
 - iii. Still failed: the response format was unclear.
 - iv. Next improvement: Specified output example.
4. Version 4:
 - a. Layer added: Specified output example.
 - b. Build the backend for an AI chatbot for university students (current, alumni, prospective). The chatbot is being developed as an AI-powered university assistant for an internship capstone project. It must support both voice and text conversations, retrieve information from the university website using Retrieval-Augmented Generation (RAG), provide citations with every response, and maintain a low response time. For example, if a student asks, "When is the admission deadline?", the response should be like: "The admission deadline is 31 August 2026"

according to the Admissions Office for the undergraduate courses.
Source: admissions.html”.

c. Notes:

- i. Changes in the prompt: the chatbot’s expected response’s structure is now defined.
- ii. Improvement in the output: Claude better understood how citations must be shown and what backend services are needed to render it properly.
- iii. Still failed: the chatbot’s response time was very long and inconsistent.
- iv. Next improvement: Quality criteria.

5. Version 5:

- a. Layer added: Quality criteria
- b. Build the backend for an AI chatbot for university students (current, alumni, prospective). The chatbot is being developed as an AI-powered university assistant for an internship capstone project. It must support both voice and text conversations, retrieve information from the university website using Retrieval-Augmented Generation (RAG), provide citations with every response, and maintain a low response time. For example, if a student asks, “When is the admission deadline?”, the response should be like: “The admission deadline for is 31 August 2026 according to the Admissions Office for the undergraduate courses. Source: admissions.html”. The backend must prioritise, low response time, accurate retrieval, transparency, minimal hallucinations and maintainable code.

c. Notes:

- i. Changes in the prompt: the measurable quality requirements are highlighted.
- ii. Improvement in the output: Claude’s response placed greater emphasis on retrieval accuracy, latency optimisation, citation reliability, scalability, and hallucination mitigation. The recommended backend architecture became more suitable for a production-ready RAG system rather than simply listing technologies.
- iii. Still failed: Claude did not provide the step-by-step roadmap before the final solution, so I am unsure of the changes and the respective justifications.
- iv. Next improvement: Step decomposition.

6. Version 6:

- a. Layer added: Step decomposition
- b. Build the backend for an AI chatbot for university students (current, alumni, prospective). The chatbot is being developed as an AI-powered university assistant for an internship capstone project. It must support

both voice and text conversations, retrieve information from the university website using Retrieval-Augmented Generation (RAG), provide citations with every response, and maintain a low response time. For example, if a student asks, “When is the admission deadline?”, the response should be like: “The admission deadline for is 31 August 2026 according to the Admissions Office for the undergraduate courses. Source: admissions.html”. The backend must prioritise, low response time, accurate retrieval, transparency, minimal hallucinations and maintainable code. Before generating the code, the system must: identify the system requirements, assess the required backend services, catalogue possible risks and the optimal route, provide possible recommendations in the working, and ask for confirmation (i.e. Am I okay to have my system developed the way Claude is suggesting?).

c. Notes:

- i. Changes in the prompt: Claude is now instructed to analyze the steps it will take as detailed in the prompt and request use confirmation to ensure the direction its taking matches the user’s intended directions.
- ii. Improvement in the output: The response became significantly more structured. Instead of immediately proposing a solution, Claude first analysed the project requirements, justified its technical decisions, identified potential risks, and then presented the recommended backend architecture. This made the reasoning process much easier to follow and increased confidence in the final solution.
- iii. Still failed: The planning stage became longer and more detailed, which increased the overall response length. However, for smaller projects, this level of analysis may not always be necessary. The increased verbosity may also confuse the designer.

Final reusable prompt:

Act as a senior AI software engineer experienced in Retrieval-Augmented Generation (RAG), Python, and LiveKit. Build the backend for an AI-powered university chatbot that supports voice and text interactions. The chatbot should retrieve information from a university website using RAG, provide citations for every response, and prioritise low latency, retrieval accuracy, transparency, maintainability, and minimal hallucinations. Before generating the solution, identify the system requirements, assess the required backend services, evaluate potential risks, recommend the most suitable architecture with justification, and ask for confirmation before producing the final implementation.